# Series and Approximations

1. Taylor Series
   - expanding a function at a point
   - power series arithmetic
2. Approximations
   - truncated power series
   - Padé approximations

MCS 320 Lecture 23
Introduction to Symbolic Computation
Jan Verschelde, 16 October 2026

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# Taylor Series

The *Taylor series* of $f(x)$ at $x = a$ is

$$f(a) + f'(a)(x-a) + f''(a)\frac{1}{2}(x-a)^2 + f'''(a)\frac{1}{3!}(x-a)^3 + \cdots$$

or written with the derivative operator $D$, up to order $n$

$$\sum_{k=0}^{n-1} \frac{D^k f(a)}{k!}(x-a)^k + O((x-a)^n),$$

where

- $D^k f(a)$ is the value of the $k$-th derivative of $f(x)$ at $x = a$,
- $O((x-a)^n)$ is the *order* of the series.

Application: the tangent line to a curve at a point is the Taylor series truncated after the linear term.

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# Power Series Arithmetic

We can multiply polynomials, but we cannot take their inverse.

Series with a nonzero constant term have a multiplicative inverse.

Example:

Given is $a = a_0 + a_1 x + a_2 x^2 + \cdots$,
compute $b = b_0 + b_1 x + b_2 x^2 + \cdots$, so $a \cdot b = 1 + O(x^3)$.

This is equivalent to solving a linear system in $b_0$, $b_1$, $b_2$:

$$\begin{aligned} a_0 b_0 &= 1, \\ a_0 b_1 + a_1 b_0 &= 0, \\ a_0 b_2 + a_1 b_1 + a_2 b_0 &= 0. \end{aligned}$$

If $a_0 \neq 0$, then $b_0 = 1/a_0$ and the system has a unique solution.

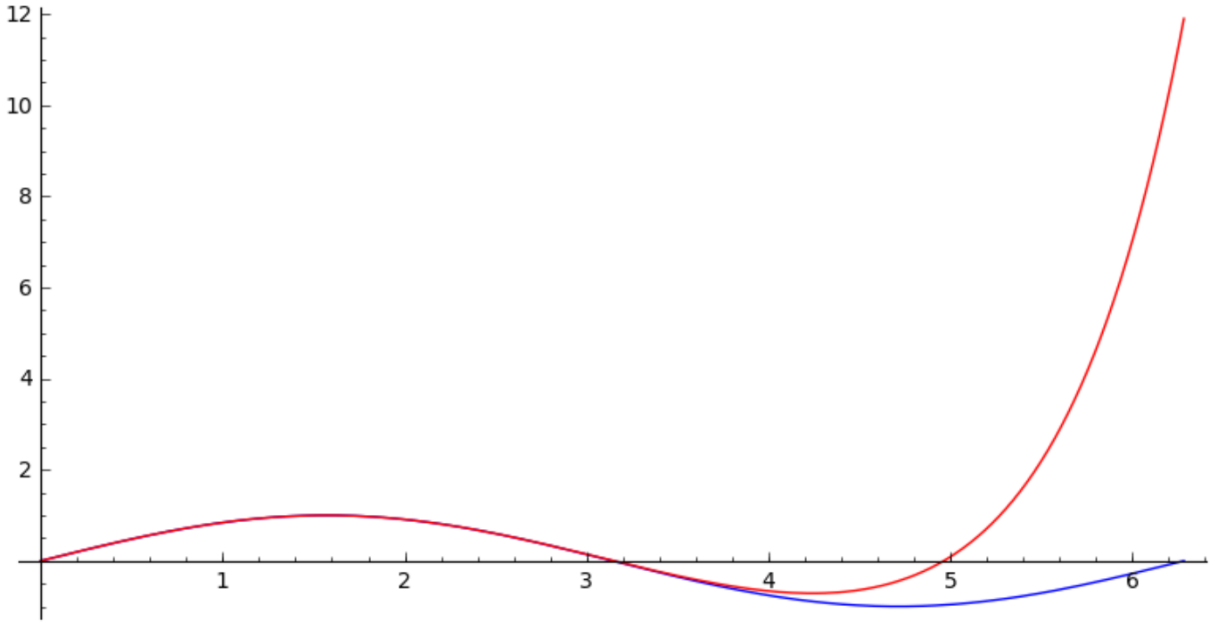
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# Truncated Power Series

Using a truncated power series to approximate $\sin(x)$:




Power series are good locally, but diverge globally.

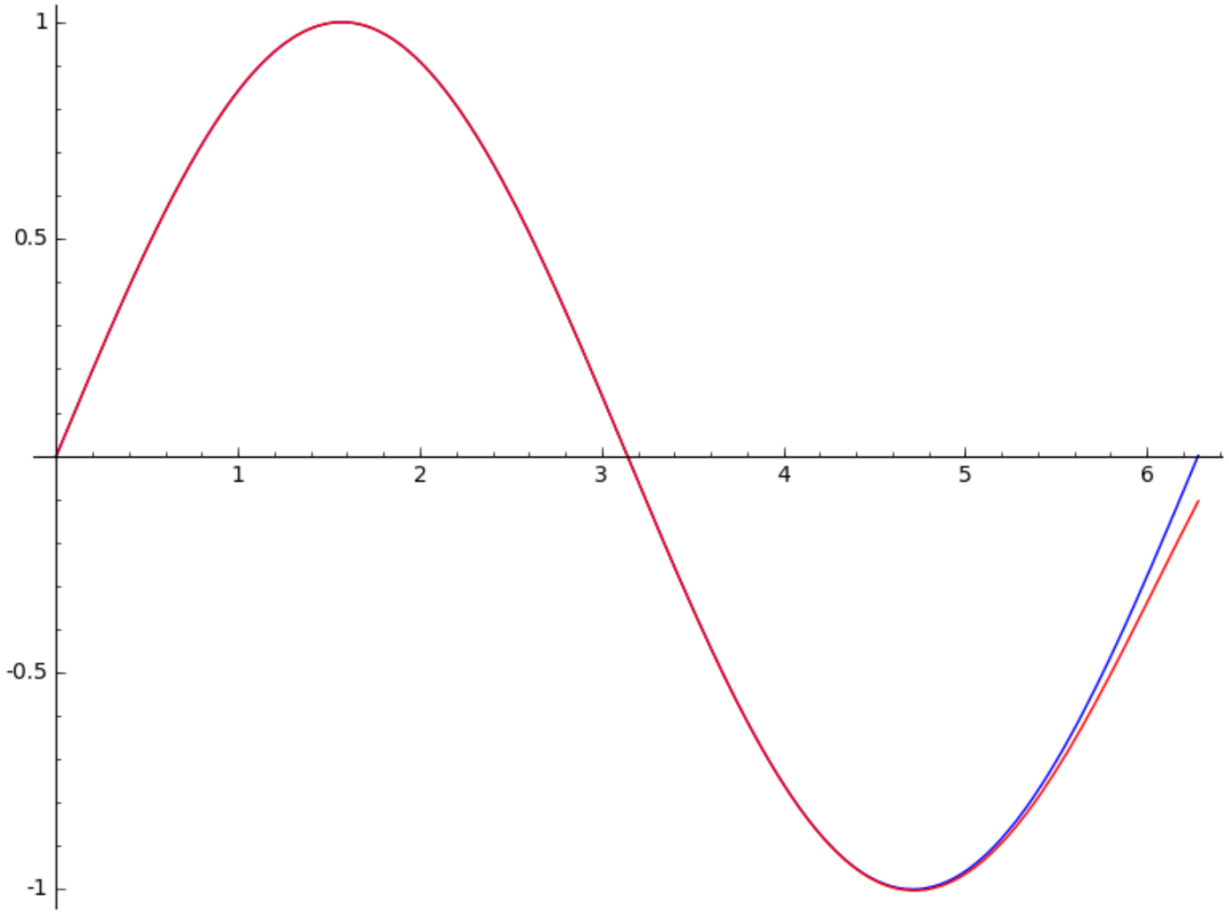
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# Padé Approximations

Using a rational approximation to approximate $\sin(x)$:




Rational approximations can achieve a global, uniform convergence.